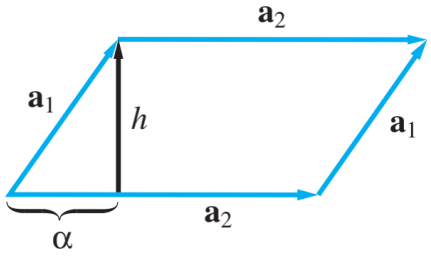
MAT 2040 Linear Algebra

Assignment 7

Please note that

- Released date: 2024/07/03
- Due: 2024/07/15
- Late submission is NOT acceptable.
- Please submit your assignment as a PDF file titled "student ID + HW7".

Question 1. Let $\boldsymbol{A}$ be a 2×2 matrix with linearly independent column vectors $\mathbf{a}_1$ and $\mathbf{a}_2$. If $\mathbf{a}_1$ and $\mathbf{a}_2$ are used to form a parallelogram P with altitude h (see the figure), show that $h^2 \|\mathbf{a}_2\|^2 = \|\mathbf{a}_1\|^2 \|\mathbf{a}_2\|^2 - \left(\mathbf{a}_1^T \mathbf{a}_2\right)^2$



Question 2. Find the constants a and b such that the vectors $\mathbf{u} = \begin{bmatrix} a \\ 4 \\ -b \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} a \\ 1 \\ b \end{bmatrix}$ are orthogonal and $a = b + 1$.

Question 3. State whether each of the following is true or false:

(a) If the subspaces V and W are orthogonal, then $V^\perp$ and $W^\perp$ are also orthogonal.

(b) If V is orthogonal to W and W orthogonal to Z then V is orthogonal to Z.

Question 4. Let $\mathbf{x}$ and $\mathbf{y}$ be linearly independent vectors in $\mathbb{R}^n$ and let $\mathcal{S} = \text{Span}\{\mathbf{x}, \mathbf{y}\}$. Construct a matrix by $\boldsymbol{A} = \mathbf{x}\mathbf{y}^T + \mathbf{y}\mathbf{x}^T$.

(a) Show that $\boldsymbol{A}$ is symmetric.

(b) Show that $\text{Null}(\boldsymbol{A}) = \mathcal{S}^\perp$.

Question 5. Find the angle between the vectors $\mathbf{v}$ and $\mathbf{w}$:

$$\mathbf{v} = (2, 1, 3)^T, \ \mathbf{w} = (6, 3, 9)^T$$

Question 6. For the following pair of vectors $\mathbf{x}$ and $\mathbf{y}$, find the vector projection $\mathbf{p}$ of $\mathbf{x}$ onto $\mathbf{y}$ and verify that $\mathbf{p}$ and $\mathbf{x} - \mathbf{p}$ are orthogonal:

$$\mathbf{x} = (3, 5)^T, \ \mathbf{y} = (1, 1)^T$$

Question 7. If $\mathbf{x} = (x_1, x_2)^T, \mathbf{y} = (y_1, y_2)^T$, and $\mathbf{z} = (z_1, z_2)^T$ are arbitrary vectors in $\mathbb{R}^2$, prove that:

(a) $\mathbf{x}^T\mathbf{x} \geq 0$

(b) $\mathbf{x}^T\mathbf{y} = \mathbf{y}^T\mathbf{x}$

(c) $\mathbf{x}^T(\mathbf{y} + \mathbf{z}) = \mathbf{x}^T\mathbf{y} + \mathbf{x}^T\mathbf{z}$

Question 8. (a) Let S be the subspace of $\mathbb{R}^3$ spanned by the vectors $\mathbf{x} = (x_1, x_2, x_3)^T$ and $\mathbf{y} = (y_1, y_2, y_3)^T$. Let

$$\boldsymbol{A} = \begin{pmatrix} x_1 & x_2 & x_3 \\ y_1 & y_2 & y_3 \end{pmatrix}.$$

Show that $S^\perp = \text{Null}(\boldsymbol{A})$.

(b) Find the orthogonal complement of the subspace of $\mathbb{R}^3$ spanned by $(1, 2, 1)^T$ and $(1, -1, 2)^T$.

Question 9. For each of the following systems $\boldsymbol{A}\mathbf{x} = \mathbf{b}$, find all least squares solutions:

(a) $\boldsymbol{A} = \begin{pmatrix} 1 & 2 \\ 2 & 4 \\ -1 & -2 \end{pmatrix}$, $\mathbf{b} = \begin{pmatrix} 3 \\ 2 \\ 1 \end{pmatrix}$

(b) $\boldsymbol{A} = \begin{pmatrix} 1 & 1 & 3 \\ -1 & 3 & 1 \\ 1 & 2 & 4 \end{pmatrix}$, $\mathbf{b} = \begin{pmatrix} -2 \\ 0 \\ 8 \end{pmatrix}$

Question 10. Let $\boldsymbol{P} = \boldsymbol{A}\left(\boldsymbol{A}^T\boldsymbol{A}\right)^{-1}\boldsymbol{A}^T$, where $\boldsymbol{A}$ is an $m \times n$ matrix of rank n.

(a) Show that $\boldsymbol{P}^2 = \boldsymbol{P}$.

(b) Prove that $\boldsymbol{P}^k = \boldsymbol{P}$ for $k = 1, 2, \ldots$

(c) Show that $\boldsymbol{P}$ is symmetric. [Hint: If $\boldsymbol{B}$ is non-singular, then $\left(\boldsymbol{B}^{-1}\right)^T = \left(\boldsymbol{B}^T\right)^{-1}$.]

Question 11. Let $\mathbf{x} = (1, 1, 1, 1)^T$ and $\mathbf{y} = (8, 2, 2, 0)^T$.

(a) Determine the angle θ between $\mathbf{x}$ and $\mathbf{y}$.

(b) Find the vector projection $\mathbf{p}$ of $\mathbf{x}$ onto $\mathbf{y}$.

(c) Verify that $\mathbf{x} - \mathbf{p}$ is orthogonal to $\mathbf{p}$.

(d) Compute $\|\mathbf{x} - \mathbf{p}\|, \|\mathbf{p}\|, \|\mathbf{x}\|$ and verify that the Pythagorean law is satisfied.

Question 12. Let $\mathbf{x}$ and $\mathbf{y}$ be linearly independent vectors in $\mathbb{R}^2$. If $\|\mathbf{x}\| = 2$ and $\|\mathbf{y}\| = 3$, what, if anything, can we conclude about the possible values of $\left|\mathbf{x}^T\mathbf{y}\right|$?

Question 13. For the following matrix, determine a basis for each of the subspaces $\text{Row}\left(A^T\right)$, $\text{Null}(A)$, $\text{Row}(A)$, and $\text{Null}\left(A^T\right)$:

$$A = \begin{pmatrix} 3 & 4 \\ 6 & 8 \end{pmatrix}$$

Question 14. Let

$$\mathbf{x} = \begin{pmatrix} 4 \\ 4 \\ -4 \\ 4 \end{pmatrix} \quad \text{and} \quad \mathbf{y} = \begin{pmatrix} 4 \\ 2 \\ 2 \\ 1 \end{pmatrix}$$

(a) Determine the angle between $\mathbf{x}$ and $\mathbf{y}$.

(b) Determine the distance between $\mathbf{x}$ and $\mathbf{y}$.

Question 15. Let $\mathbf{x}$ and $\mathbf{y}$ be vectors in $\mathbb{R}^n$ and define

$$\mathbf{p} = \frac{\mathbf{x}^T\mathbf{y}}{\mathbf{y}^T\mathbf{y}}\mathbf{y} \quad \text{and} \quad \mathbf{z} = \mathbf{x} - \mathbf{p}$$

(a) Show that $\mathbf{p} \perp \mathbf{z}$. Thus, $\mathbf{p}$ is the vector projection of $\mathbf{x}$ onto $\mathbf{y}$; that is, $\mathbf{x} = \mathbf{p}+\mathbf{z}$, where $\mathbf{p}$ and $\mathbf{z}$ are orthogonal components of $\mathbf{x}$, and $\mathbf{p}$ is a scalar multiple of $\mathbf{y}$.

(b) If $\|\mathbf{p}\| = 6$ and $\|\mathbf{z}\| = 8$, determine the value of $\|\mathbf{x}\|$.

Question 16. Let S be the subspace of $\mathbb{R}^n$ spanned by the vectors $\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_k$. Show that $\mathbf{y} \in S^\perp$ if and only if $\mathbf{y} \perp \mathbf{x}_i$ for $i = 1, \ldots, k$.